

What is Cloud Computing? Definition(s)

“A standardized IT capability (services, software, or infrastructure) delivered via internet technologies in a pay-per-use, self-service way.”

“Cloud computing is a model for enabling ubiquitous, convenient, on-demand access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.”

Ubiquitous: able to access cloud environment from anywhere.

Business Drivers (Why use Cloud computing?)

- Capacity Planning: determine and fulfil future demands of an organization's IT resources.
 - Computing Resources
 - CPU
 - Memory
 - Storage
 - Network
 - Over-provisioning
 - Under-provisioning
- Cost Reduction
- BCP/DR
 - RTO (Recovery Time Objective)
 - RPO (Recovery Point Objective)
- Organization Agility
 - Responsiveness to change

Basic concepts and Terminologies

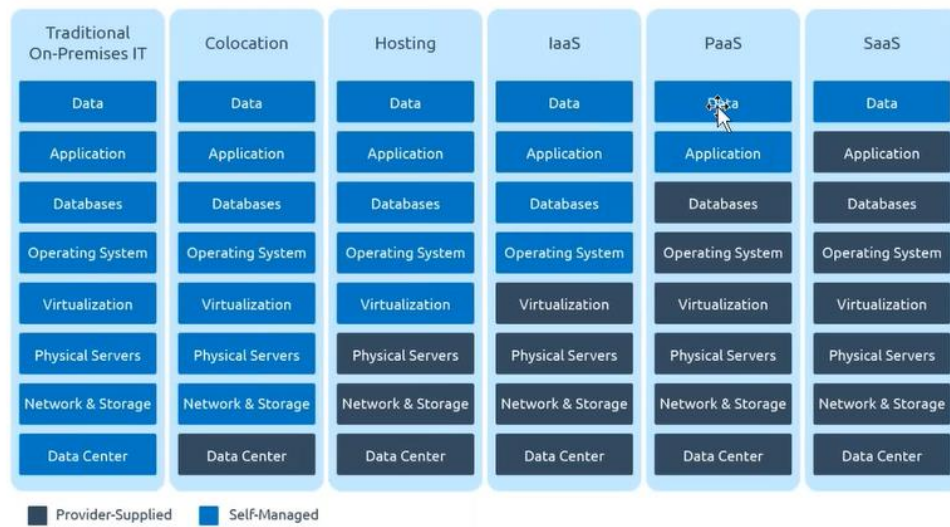
- Cloud
 - Distinct Distributed IT environment that is designed for the purpose of remotely provisioning scalable and measurable IT resources.
- IT Resource
- On Premise
- Cloud Provider: (AWS, GCP, Azure.....)

- Cloud Consumer
- Scaling (towards service)
 - Horizontal: (Scaling out/ Scaling In)
 - Vertical: Requires downtime (Scaling Up/ Scaling down)

Cloud Characteristics

- On-demand usage
- Ubiquitous usage
 - Accessible from any where
 - Accessible from any device
- Multitenancy
 - Pools: Physical servers, virtual server, storage, network, memory etc.
- Elasticity (IT resources)
 - Ability to transparently scale IT resources as required in response to runtime conditions pre-defined by cloud consumer.
- Measured usage
- Resiliency – Address reliability and availability of applications
 - Fault tolerance
 - Failover technique readily available
- BCP/DR inbuilt in the cloud platform.

Cloud Delivery Models

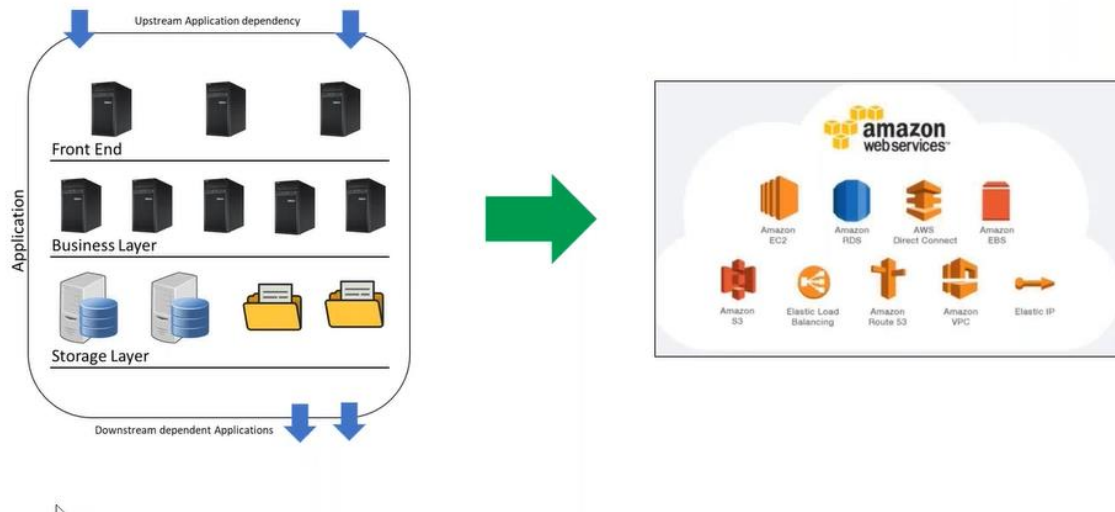


- IaaS
- PaaS
- SaaS
- Storage as a Service
- Database as a Service
- Security as a Service
- Communication as a Service
- Integration as a Service
- Process as a Service
- Function as a Service

Cloud Deployment Models

- Public Cloud
- Community Cloud
- Private
- Hybrid Cloud

Migration from on-premise to AWS cloud

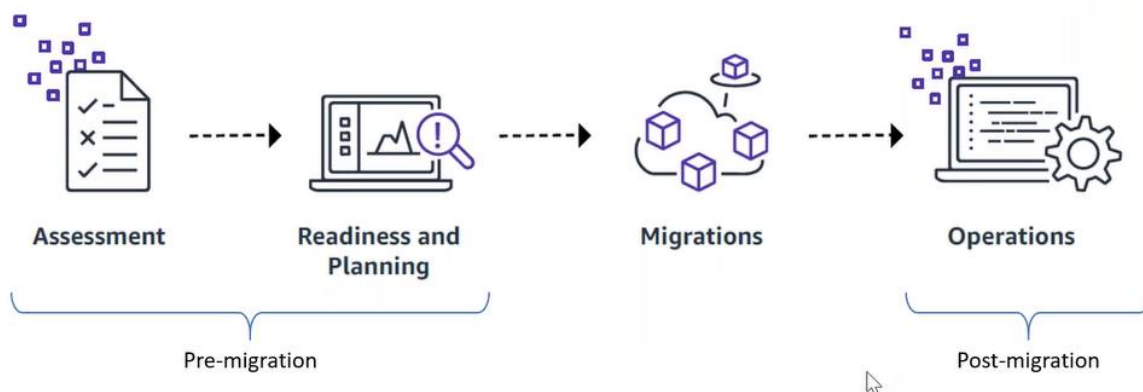


H/W never migrated to the cloud, S/W (O/S, app) installed on top of that migrated to the cloud, and accessed over the internet.

Upstream Application Dependency- Input to application

Downstream Application Dependency- output of application

Migration Phases



- Pre-Migration (70%)
 - Assessment
 - Readiness and Planning
- Migration (20%)
- Post-Migration (10%)

- Operations

Assessment:



• Assessment

- What applications do we have
- Which application(s) be migrated to cloud environment
- People, Process and Technology
 - Gaps in existing setup
 - Changes required for Migration and Post migration
- TCO – current and post migration

TCO: total cost of ownership.

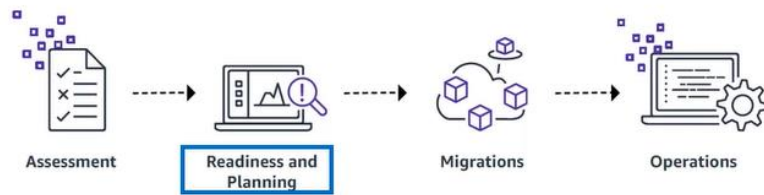
Storage, Networking, Computing

Typical Application Artifacts

- **Application (focus of migration)**
 - **Front End Components**
 - Server(s) – Optimized for specific application
 - Third party component(s)- optional
 - **Business Layer Components**
 - Application specific logic.
 - Hosting H/W specific
 - Third party component(s)- optional
 - **Storage Layer Components**
 - File Storage: NFS, SMB
 - Data Storage: logs, temp etc.
 - Database: media(s)
 - **Integration Components**
 - One or more dependencies.
 - via API-online
 - Batch mode- offline
 - **Network Components**

- Typically, not migrate.
- **Process Assessment**
 - End to End application process to identify challenges
 - Run through all use cases e.g. from where input comes from? What to do with those inputs? What is the end result of those inputs?
 - Batch process
 - Sources data (What type of the data coming from)
 - ETL
 - Frequency
 - Third party process and their support during migration
 - Reporting requirements (who is the consumer of those reports)
 - Manual processes involved if any
 - Communication modes
 - New Release roll-out
- **Technology Assessment**
 - **Operating System**
 - Constraints
 - Special Requirements
 - **Patch Management**
 - **New Release roll-outs**
 - **Back-ups/Restore**
 - **BCP/DR**

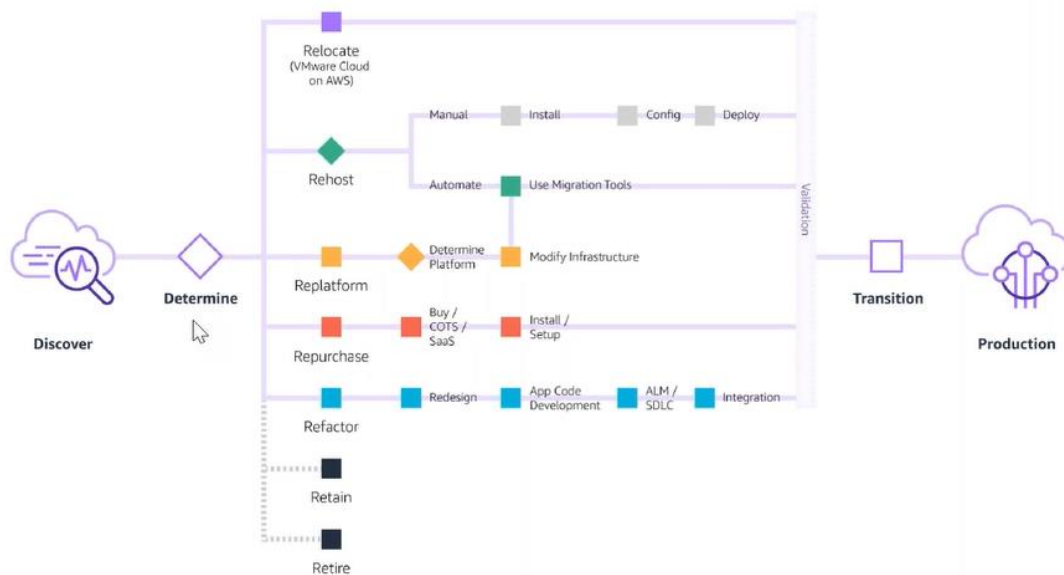
Readiness and Planning



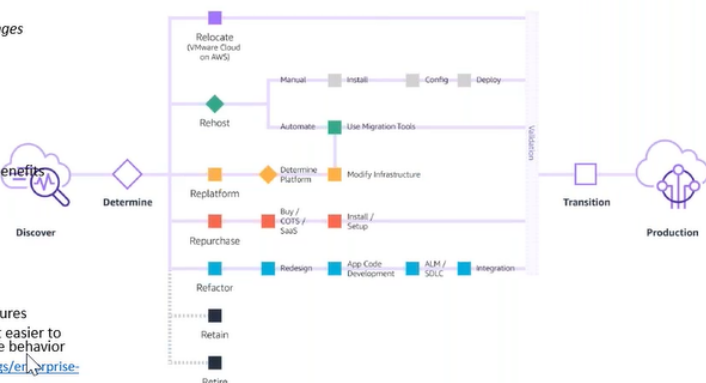
- **Readiness & Planning**

- Address the gaps identified in the assessment phase
- Build your baseline environment – technology and process
- Develop cloud skills - people
- Creating a strong migration plan
 - Migrating application priorities
 - Use 7 R's as migration strategy for each application
- Identify tools for migrating different application components
- Evaluate direct connection options to AWS cloud for initial migration or ongoing operations

Common Application Migration Strategy (7'R)

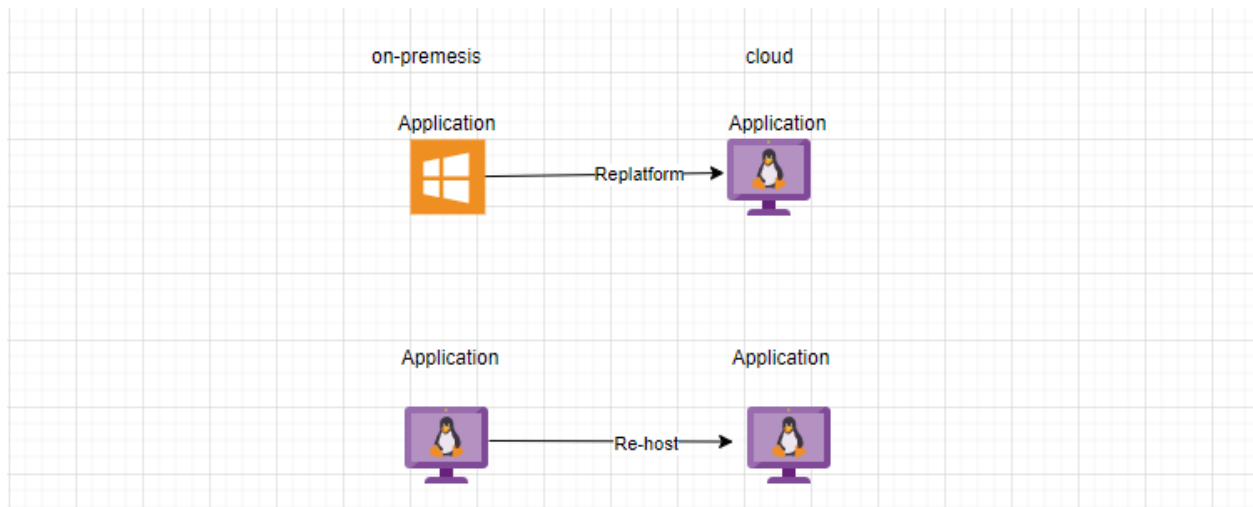


- **Relocate**
 - Move vSphere-based applications to AWS without application changes
- **Re-Hosting**
 - Just rehost existing application for quick scaling
 - aka *Lift-and-Shift*
- **Re-Platforming**
 - Core architecture remains intact
 - Minor to medium optimizations or tweaking to leverage tangible benefits
 - aka *Lift-Tinker-and-Shift*
- **Re-Purchasing**
 - Replacing existing capability via a different product
 - e.g. CRM system moved to Salesforce.com
- **Re-Factoring / Re-Architecting**
 - Revisiting application Architecture using cloud native services/features
 - A change made to the internal structure of the software to make it easier to understand and cheaper to modify without changing its observable behavior
 - Read this article for more insight - <https://aws.amazon.com/blogs/enterprise-strategy/the-great-cloud-refactoring-debate/>
- **Retire**
 - Not required any more, sunset.
- **Retain**
 - In a rare situation application may not be ready or not required to migrate for now



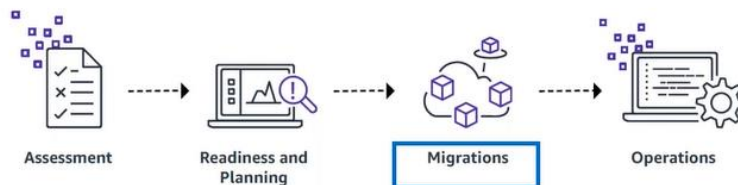
Re-Host: Windows<> Windows

Re-Platform: Windows<>Linux (Underneath platform change)



Interpreter, well know your on-prem (one language) as well as Cloud (other language), there is always exam of language not the interpreter.

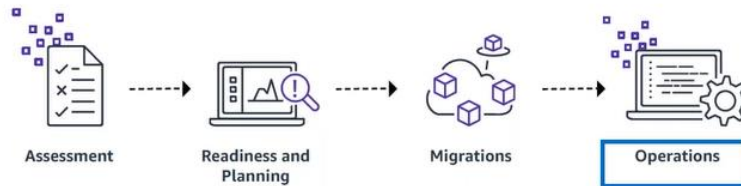
Migration Phase:



• Actual Migration Process

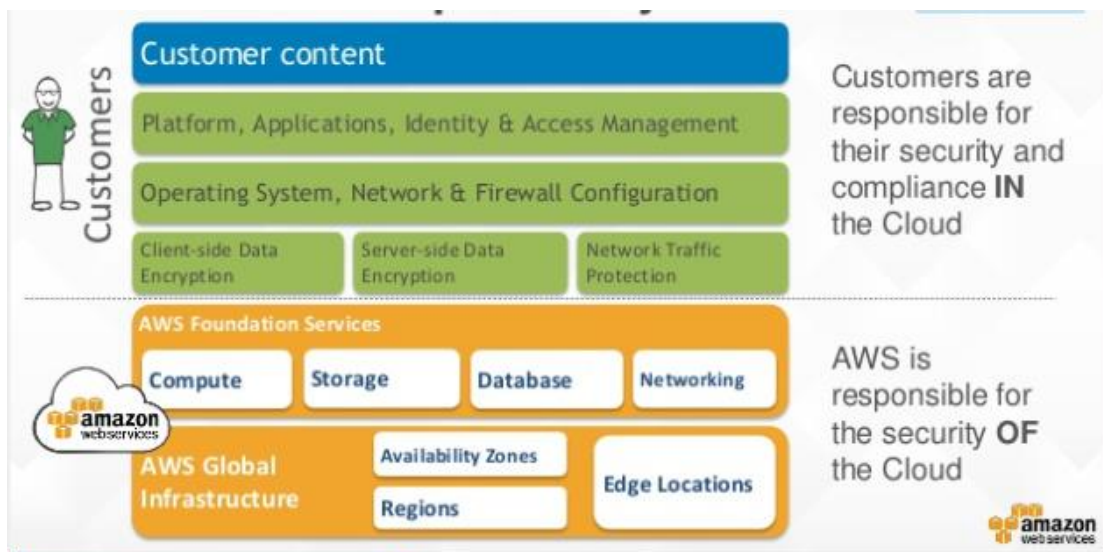
- Application designed, migrated and validated as per 7 R's strategy
- Pre-migration preparation
 - Logistics around data collection
 - Encryption/Decryption of existing data
 - Existing encrypted data need to be decrypted before presenting it to migrating tool, re-encrypt post migration – there can be exceptions to this
 - Checklist for migration tracking
- Post-migration activities
 - Validation in context of completeness, correctness and integrity of the migrated component

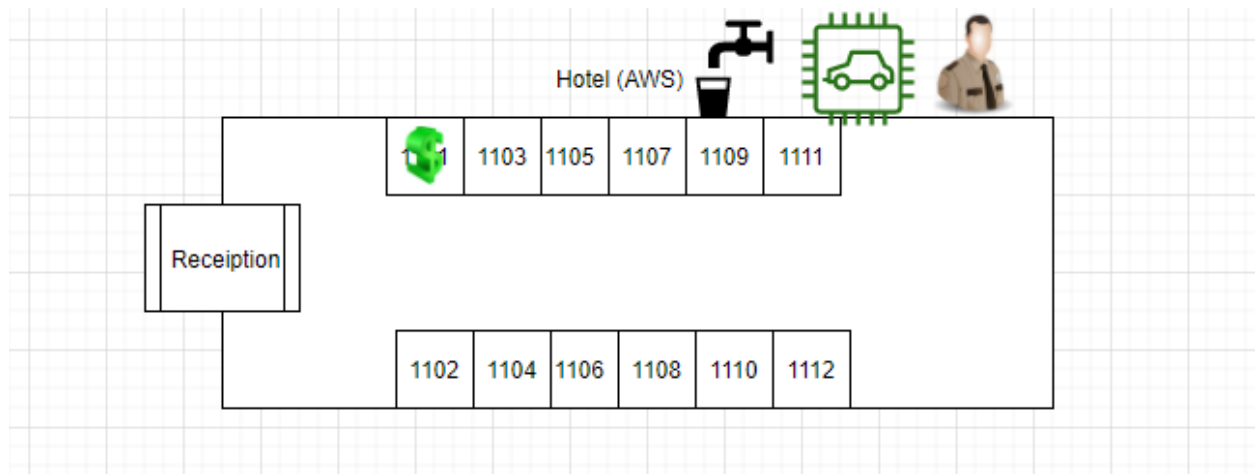
Operations:



- Operations – Post Migration
 - Plan to turn off old systems
 - Establish two way sync process to hybrid setup
 - Identify opportunities to automate
 - Change requests
 - Monitoring
 - Patch Management
 - Security
 - Backups
 - Continuous Improvement mindset

Shared Responsibility Model





Internet provisioning is AWS responsibility, securing the same is our responsibility.

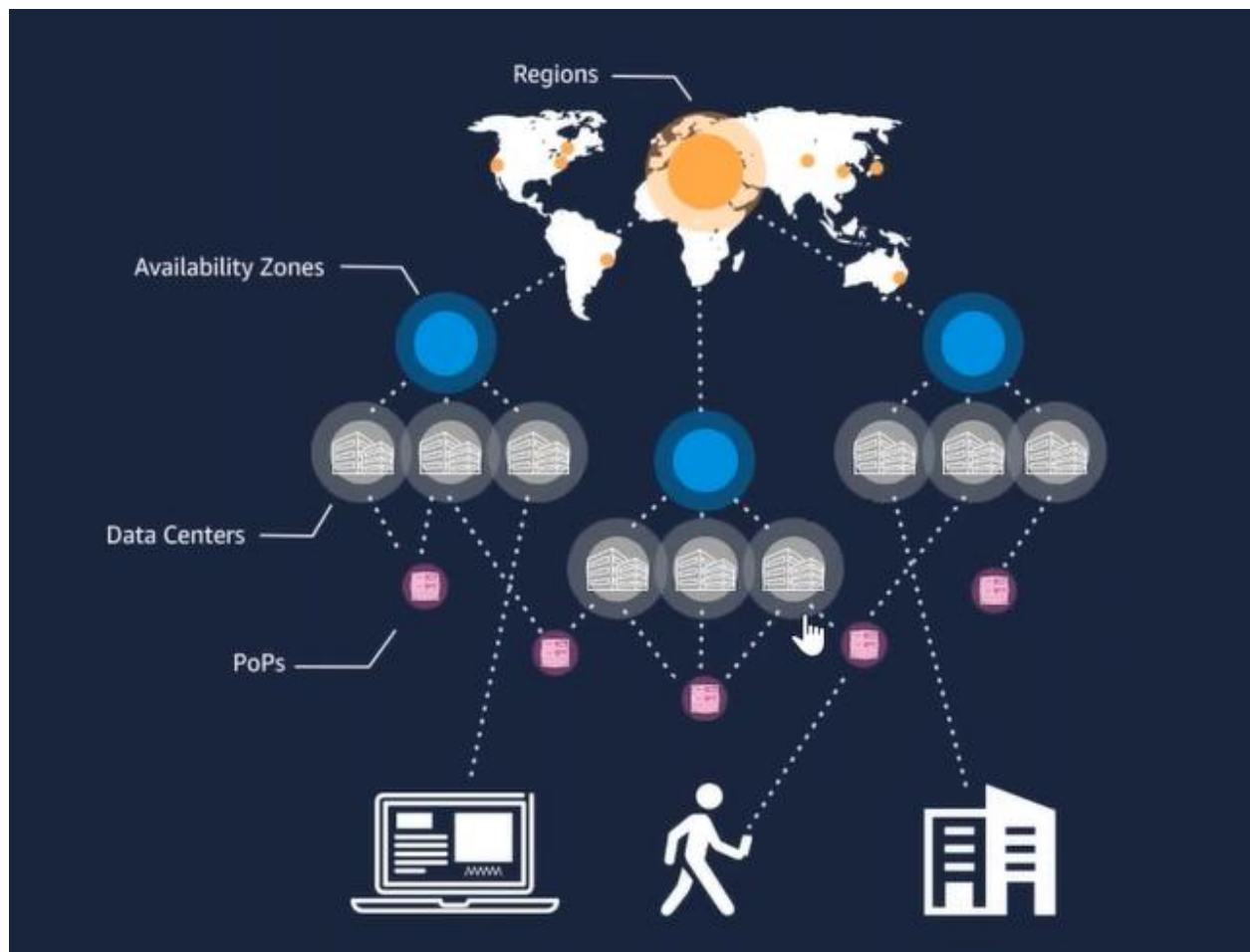
How O/S responsibility lies on Customer?

AWS just provide the O/S you are responsible for apply patches.

Region Failure?

It's our responsibility to design as per disaster recovery Using Route53

<https://infrastructure.aws/>



- Gov Cloud: Not accessible from public
- Beijing: China Government
- Generally Available

Partition (Different Networks):

- aws
- aws-us-gov
- aws-cn
- Regions: Geographical Location (Logical term)
- Availability Zone: (Logical term)
 - Data Centers (Physical presence)
- Local Zone: Reduce the latency (full fleshed data center)

- Points of Presence: (minimum 10GB lines connected with data centers) also called Edge Location. Just for caching purpose. (Wave Length).

status.aws.com (AWS service status)

- personal health dashboard (Is AWS down service impact in my account?)
 - Event Log

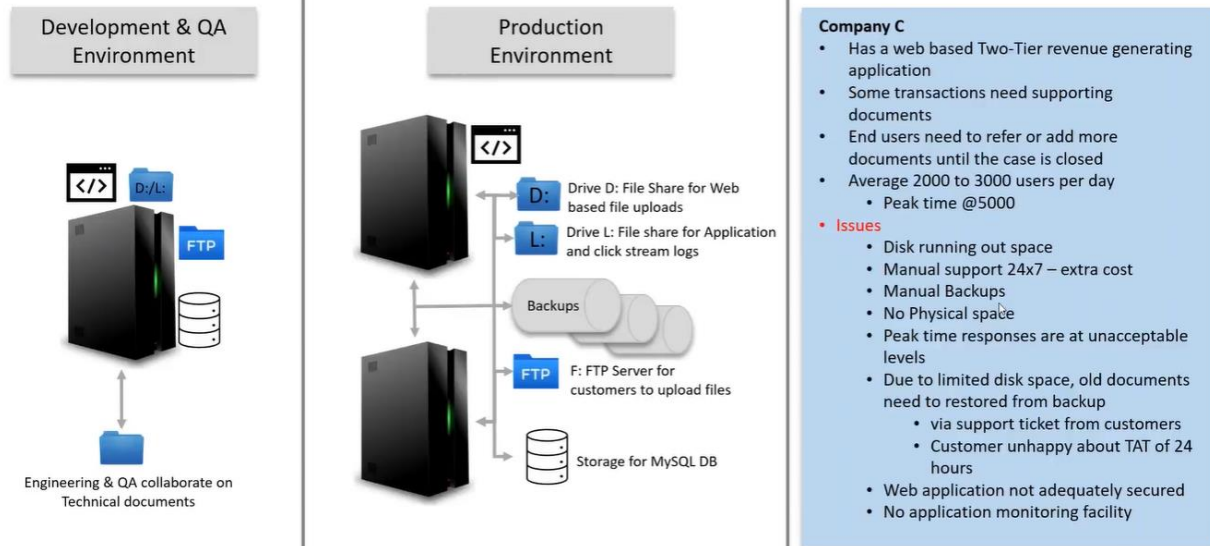
arn (amazon resource name):

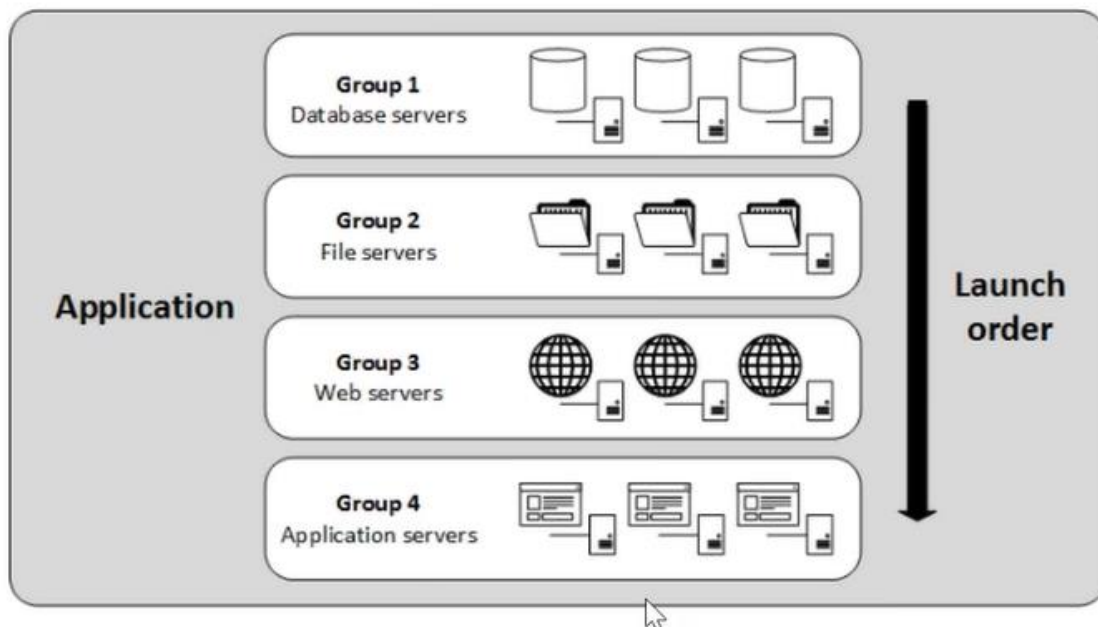
partition:service::region:accountid:resource-id

- cloudfare.com

Use Case - Company C

edureka!





Total Cost Projection:

Always we have to provide the range not the actual estimation.